

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: T Y B Tech Branch : Chemical Engineering **Semester :** VI

Subject Code & Name: Chemical Reaction Engineering II (BTCHC 601)

Max Marks: 60

Date: 18/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The first order series reaction $A \rightarrow B \rightarrow C$ is conducted in a batch reactor. The initial concentration of A, B and C (C_{AO} , C_{BO} , C_{CO} respectively) are all non-zero. The variation of CB with reaction time will not show a maximum, if	4/CO1	1
	a) $K_2 C_{BO} \gg K_1 C_{AO}$ b) $K_2 C_{BO} \ll K_1 C_{AO}$ c) $C_{BO} \gg C_{AO}$ d) $C_{BO} \ll C_{AO}$		
2	For series reaction, the relative yield	5/CO1	1
	a. Is always greater for plug-flow reactor than for the single CSTR of the same volume b. Statement in (A) is wrong c. Decreases with increasing conversion d. Both (A) and (C) hold good		
3	The Monod Model relates	2/CO2	1
	a. greater affinities to substrate b. lower affinities to substrate c. greater affinities to substrate d. greater affinities to substrate		
4	Enzymes	1/CO2	1
	(a) used to increase the activation (b) used to convert inorganic (c) used to change the product (d) found in organism		

	energy of reaction	catalyst to organic catalyst	formation in a desired manner			
5	In a batch reactor specific growth rate				5/CO2	1
	a. remains constant throughout the process	b. increases exponentially	c. changes during the fermentation	d. is highest during the stationary phase		
6	For shrinking core model which resistance is variable				2/CO3	1
	a. Resistance offered by gas layer	b. Resistance offered by ash layer	c. Resistance of chemical reaction	d. Resistance offered by turbulent gas-phase		
7	In which non-catalytic gas-solid reaction model fractional conversion of solid is proportional to reaction time				2/CO3	1
	(a) Ash layer control diffusion	(b) Chemical reaction control process	(c) Gas film-controlled diffusion	(d) None of the above		
8	For an ideal mixed flow reactor (CSTR), the exit age distribution $E(t)$ is given by				2/CO4	1
	a. dirac delta function	b. step function	c. ramp function	d. None of the above		
9	Residence time distribution (RTD) of a reactor is independent of				2/CO4	1
	a. micro mixing	b. volume of reactor	c. height of the reactor	d. width of the reactor		
10	The 'E' curve for a non-ideal reactor defines the fraction of fluid having age between t and $t + dt$				3/CO4	1
	a. At the inlet	b. At the outlet	c. in the reactor	d. Averaged over the inlet and outlet		
11	The single parameter model proposed for describing non-ideal flow is the _____ model.				2/CO5	1
	a. Tank in	b. Dispersion	c. Both A & B	d. Neither A		

	series			nor B		
12	In an ideal tubular-flow reactor				2/CO4	1
	a. There is no mixing in longitudinal direction	b. Mixing takes place in radial direction	c. There is a uniform velocity across the radius	d. all of the above		
Q. 2	Solve Any Two of the following.					12
A)	Discuss the algorithms for multiple reactions where at least one species is involved in only one reaction.				1/CO1	6
B)	Explain the Stoichiometric Table using fractional Conversion for Multiple Reactions.				2/CO1	6
C)	Discuss the choice of the reactor and conditions to Maximize the desired products for the following parallel reactions i) $A + B \rightarrow D \quad r_D = k_1 C_A^{a_1} C_B^{b_1}$ ii) $A + B \rightarrow U \quad r_U = k_2 C_A^{a_2} C_B^{b_2}$ Consider all possible combinations of reactions orders and select the reaction scheme that will maximize $S_{D/U}$				2/CO1	6
Q.3	Solve Any Two the following.					12
A)	Define and discuss PSSH. Explain different types of intermediates with examples				1/CO2	6
B)	Propose a mechanism for the decomposition of Nitrous oxide ($N_2O \rightarrow N_2 + \frac{1}{2} O_2$) and show that the rate is given by: $-r_{N_2O} = k_1 [N_2O]^2 / \{1 + k' N_2O\}$				3/CO2	6
C)	Derive Michaelis-Menten kinetic equation for the enzyme catalyzed reaction $A \rightarrow R$				3/CO2	6
Q. 4	Solve Any Two of the following.					12
A)	An ore of uniform size particles is to be roasted in a fluidized bed reactor. The time required for complete conversion of solid particles is 20 min and the mean residence time of particles in the bed is 48 min. the solids remain unchanged in size during reaction. Calculate the fraction of the original ore remaining unconverted assuming (i) the				3/CO3	6

	chemical reaction step to be rate controlling (ii) the ash diffusion step to be rate controlling																														
B)	Discuss the effects of temperature, time and particle size on determination of rate controlling step for fluid particle reactions	2/CO3	6																												
C)	Explain progressive conversion model for spherical particles of changing size.	2/CO3	6																												
Q.5	Solve Any Two of the following.		12																												
A)	Discuss about various types of inputs for tracer study with sketches and types of nonideality exists in non ideal flow reactors	1/CO4	6																												
B)	State and discuss with significance, the various dimensionless RTD parameters	1/CO4	6																												
C)	The concentration readings given below represent a continuous response to pulse input into a closed <table><tr><td>Time in min</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td></tr><tr><td>Effluent conc .g /lit</td><td>0</td><td>3</td><td>5</td><td>5</td><td>4</td><td>2</td><td>1</td><td>0</td></tr></table> a) Plot C, E and F curve b) Calculate mean residence time c) Calculate variance and standard deviation	Time in min	10	20	30	40	50	60	70	80	Effluent conc .g /lit	0	3	5	5	4	2	1	0	3/CO4	6										
Time in min	10	20	30	40	50	60	70	80																							
Effluent conc .g /lit	0	3	5	5	4	2	1	0																							
Q. 6	Solve Any Two of the following.		12																												
A)	Discuss and derive the axial dispersion model to describe the non-ideal behavior in tubular flow reactor.	1/CO5	6																												
B)	Discuss the classification of models for non ideal reactors ?	1/CO5	6																												
C)	The first order reaction $A \rightarrow B$ is carried out in a 10cm-diameter tubular reactor 6.36 m in length. The specific reaction rate is 0.25 min^{-1}. The results of a tracer test carried out on this reactor are shown in Table. <table><tr><td>Time in min</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>12</td><td>14</td></tr><tr><td>Effluent conc .g /lit</td><td>0</td><td>1</td><td>5</td><td>8</td><td>10</td><td>8</td><td>6</td><td>4</td><td>3</td><td>2.2</td><td>1.5</td><td>0.6</td><td>0</td></tr></table> Calculate conversion using (a) the closed vessel dispersion model, (b) PFR, (c) the tanks-in-series model, and (d) a single CSTR.	Time in min	0	1	2	3	4	5	6	7	8	9	10	12	14	Effluent conc .g /lit	0	1	5	8	10	8	6	4	3	2.2	1.5	0.6	0	3/CO5	6
Time in min	0	1	2	3	4	5	6	7	8	9	10	12	14																		
Effluent conc .g /lit	0	1	5	8	10	8	6	4	3	2.2	1.5	0.6	0																		
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